MSG No. 198.11 P_I2_13 [Type IV, cubic]

Table 1: Wyckoff site: `8a`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\, x,\, x]$	`[1,5,6]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, -x\right]$	`[2,10,11]`
3	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	`[3,7,12]`
4	$\left[\frac{1}{2}-x,\, -x,\, x+\frac{1}{2}\right]$	`[4,8,9]`
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, x+\frac{1}{2}\right]$	`[13,17,18]`
6	$\left[x,\, -x,\, \frac{1}{2}-x\right]$	`[14,22,23]`
7	$\left[\frac{1}{2}-x,\, x,\, -x\right]$	`[15,19,24]`
8	$\left[-x,\, \frac{1}{2}-x,\, x\right]$	`[16,20,21]`

Table 2: Wyckoff site: `12b`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{4}\right]$	`[1,14]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{3}{4}\right]$	`[2,13]`
3	$\left[-x,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[3,16]`
4	$\left[\frac{1}{2}-x,\, 0,\, \frac{3}{4}\right]$	`[4,15]`
5	$\left[\frac{1}{4},\, x,\, 0\right]$	`[5,24]`
6	$\left[0,\, \frac{1}{4},\, x\right]$	`[6,21]`
7	$\left[0,\, \frac{3}{4},\, \frac{1}{2}-x\right]$	`[7,23]`
8	$\left[\frac{1}{4},\, -x,\, \frac{1}{2}\right]$	`[8,22]`
9	$\left[\frac{1}{2},\, \frac{3}{4},\, x+\frac{1}{2}\right]$	`[9,18]`
10	$\left[\frac{3}{4},\, \frac{1}{2}-x,\, 0\right]$	`[10,20]`
11	$\left[\frac{1}{2},\, \frac{1}{4},\, -x\right]$	`[11,19]`
12	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[12,17]`

Table 3: Wyckoff site: `24c`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	`[2]`
3	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	`[4]`
5	$[z,\, x,\, y]$	`[5]`
6	$[y,\, z,\, x]$	`[6]`
7	$\left[-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	`[7]`
8	$\left[\frac{1}{2}-z,\, -x,\, y+\frac{1}{2}\right]$	`[8]`
9	$\left[\frac{1}{2}-y,\, -z,\, x+\frac{1}{2}\right]$	`[9]`
10	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, -y\right]$	`[10]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, -x\right]$	`[11]`
12	$\left[-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	`[12]`
13	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[13]`
14	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	`[14]`

continued ...

Table 3

No.	position	mapping
15	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[15]
16	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	[16]
17	$\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[17]
18	$\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[18]
19	$\left[\frac{1}{2}-y,\, z,\, -x\right]$	[19]
20	$\left[-z,\, \frac{1}{2}-x,\, y\right]$	[20]
21	$\left[-y,\, \frac{1}{2}-z,\, x\right]$	[21]
22	$\left[z,\, -x,\, \frac{1}{2}-y\right]$	[22]
23	$\left[y,\, -z,\, \frac{1}{2}-x\right]$	[23]
24	$\left[\frac{1}{2}-z,\, x,\, -y\right]$	[24]